

A One-Sided Oracle Tolerance for Lim's Smoothness-Minimizing Estimator

A repair note for Problem (B)

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Abstract

Lim's Problem (B) estimates a regression function by minimizing Sobolev roughness subject to an empirical residual constraint. Lim observed that the same empirical L_2 rate proved for Problem (A) would also hold for Problem (B) if a certain one-sided inequality, displayed as (13) in her paper, could be forced by a condition on the tolerance S_n . This note gives a rigorous repair of a proposed argument for that issue.

The main point is that the sign of the tolerance error is essential. Let

$$V_n = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2, \quad r_n = n^{-m/(2m+d)}.$$

If $S_n \leq V_n$, then Lim's inequality (13) follows immediately from the feasibility constraint. If, in addition, the undershoot is no larger than the squared minimax scale,

$$0 \leq V_n - S_n = O_p(r_n^2),$$

then, under a standard low-frequency projection condition verified below for uniform designs and Gaussian errors, the roughness of the solution remains bounded in probability. Consequently

$$\|\hat{g}_n - f^*\|_n = O_p(r_n), \quad \|\hat{g}_n - f^*\|_{L_2(P_X)} = O_p(r_n),$$

and, for $0 \leq |\alpha| \leq m - 1$,

$$\|D^\alpha(\hat{g}_n - f^*)\|_{L_2(P_X)} = O_p\left(n^{-(m-|\alpha|)/(2m+d)}\right).$$

In particular, the exact oracle choice $S_n = V_n$ is a complete oracle solution to Lim's rate question for Problem (B). It is not, by itself, a data-driven solution in the ordinary one-observation-per-design model, because V_n depends on the unobserved errors. We also record a simple obstruction showing that deterministic tolerances centered at σ^2 cannot force the needed one-sided inequality at the squared minimax scale.

1 Status and relation to Lim's question

We observe independent data

$$Y_i = f^*(X_i) + \varepsilon_i, \quad i = 1, \dots, n,$$

where X_i take values in a compact rectangle $\Omega \subset \mathbb{R}^d$. Lim's smoothness-minimizing estimator is any solution of

$$\hat{g}_n \in \arg \min_{f \in \mathcal{F}_m} J(f) \quad \text{subject to} \quad E_n(f) := \frac{1}{n} \sum_{i=1}^n \{Y_i - f(X_i)\}^2 \leq S_n, \quad (1)$$

where

$$J(f) = \sum_{|\alpha|=m} \int_{\mathbb{R}^d} \{D^\alpha f(x)\}^2 dx.$$

The issue is Lim's Remark 4: the rate proof for Problem (A) transfers to $\hat{g}_n$ if

$$\frac{1}{n} \sum_{i=1}^n \{\hat{g}_n(X_i) - f^*(X_i)\}^2 \leq \frac{2}{n} \sum_{i=1}^n \varepsilon_i \{\hat{g}_n(X_i) - f^*(X_i)\} \quad (2)$$

for all sufficiently large n . Lim wrote that a suitable condition on S_n implying this display was not known.

The repair in this note is deliberately one-sided. A two-sided condition such as

$$S_n = V_n + O_p(r_n^2)$$

does not imply (2); the positive part $S_n - V_n$ appears with the wrong sign. The condition that directly implies (2) is instead

$$S_n \leq V_n, \quad V_n := \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2. \quad (3)$$

The exact oracle tolerance $S_n = V_n$ also makes f^* feasible, hence immediately bounds $J(\hat{g}_n)$ by $J(f^*)$. This gives a clean oracle solution. The more general band

$$0 \leq V_n - S_n = O_p(r_n^2) \quad (4)$$

requires a nontrivial feasible-correction argument; Section 5 proves it under additional regularity assumptions.

Thus the answer obtained here is best described as follows.

- It is a complete solution to the literal mathematical rate question if oracle tolerances depending on the realized noise level are allowed; $S_n = V_n$ is the cleanest choice.
- It is partial progress, not a fully data-driven solution, for the practical problem of choosing S_n from ordinary data without knowing f^* or the errors.

2 Standing assumptions and standard empirical bounds

The rate statements use the following assumptions, which are essentially Lim's Assumptions 2–4 with a slightly explicit Sobolev-domain formulation.

Assumption 2.1 (Basic regression and Sobolev assumptions). *Let $m, d \in \mathbb{N}$ satisfy $2m > d$. Let $\Omega = [a, b]^d$. The design variables $X, X_1, X_2, \dots$ are iid with density τ on Ω satisfying*

$$0 < \tau_- \leq \tau(x) \leq \tau_+ < \infty.$$

The regression function satisfies $f^ \in \mathcal{F}_m$. The errors satisfy*

$$\mathbb{E}(\varepsilon_i \mid X_i) = 0, \quad \mathbb{E}(\varepsilon_i^2 \mid X_i) = \sigma^2 < \infty,$$

and are uniformly sub-Gaussian. The solution (1) is taken on the event where Problem (B) exists; under Lim's unisolvency result this event has probability tending to one.

Write

$$\|h\|_n^2 = \frac{1}{n} \sum_{i=1}^n h(X_i)^2, \quad \|h\|_P^2 = \mathbb{E}\{h(X)^2\}.$$

The following lemma is the empirical-process ingredient needed below. It is a standard localized Sobolev bound. The finite-dimensional polynomial null space of J is explicitly accounted for; omitting it is exactly the flaw in the proposed proof that motivated this repair note.

Lemma 2.2 (Localized Sobolev empirical bounds). *Assume Assumption 2.1. Fix $M < \infty$ and define*

$$\mathcal{H}_M = \{h = f - f^* : f \in \mathcal{F}_m, J(f) \leq M\}.$$

There are random variables $Z_n = O_p(1)$ and constants $C_M < \infty$ such that, with probability tending to one, every $h \in \mathcal{H}_M$ satisfies

$$|\mathbb{P}_n \varepsilon h| \leq Z_n \left[n^{-1/2} \|h\|_n + n^{-1/2} \{\|h\|_n + r_n\}^{1-d/(2m)} \right], \quad (5)$$

where $r_n = n^{-m/(2m+d)}$, and

$$\|h\|_P \leq C_M \{\|h\|_n + r_n\}. \quad (6)$$

Proof sketch with the null space included. Let $\mathcal{P}_{m-1}$ denote the polynomials of degree at most $m-1$, the null space of J . Decompose $h = p + v$, where $p \in \mathcal{P}_{m-1}$ and v is chosen in a fixed closed complement, for instance by imposing the usual moment conditions against $\mathcal{P}_{m-1}$ on Ω . On that complement, a Poincare inequality gives

$$\|v\|_{W^{m,2}(\Omega)} \leq C \{J(v)^{1/2} + \|v\|_{L_2(\Omega)}\}.$$

The class of such v with $J(v) \leq C_M$ has Birman–Solomyak entropy exponent d/m in $L_\infty(\Omega)$ and hence in every empirical metric. The polynomial part is finite dimensional; on the event that the random design is eventually unisolvent and the empirical Gram matrix on $\mathcal{P}_{m-1}$ is nonsingular, empirical and population norms on $\mathcal{P}_{m-1}$ are equivalent. Therefore the polynomial contribution to the multiplier process is at most $O_p(n^{-1/2} \|p\|_n)$, while the non-polynomial contribution is controlled by the standard sub-Gaussian multiplier maximal inequality with entropy exponent d/m ; see, for example, van de Geer [4, Lemma 8.4] together with the Birman–Solomyak entropy estimate [1].

Combining the two parts and absorbing the finite-dimensional norm equivalences gives (5). Applying the same entropy estimate to the quadratic empirical process $h \mapsto \mathbb{P}_n h^2 - Ph^2$, again treating $\mathcal{P}_{m-1}$ separately, gives (6). This is the standard “empirical-to-population” localization for Sobolev balls when $2m > d$. $\square$

3 A rate theorem once the one-sided inequality and roughness are known

Proposition 3.1 (Rate transfer for Problem (B)). *Assume Assumption 2.1. Suppose that, with probability tending to one,*

$$J(\hat{g}_n) \leq M \quad (7)$$

for some deterministic $M < \infty$, and that Lim’s one-sided inequality (2) holds. Then

$$\|\hat{g}_n - f^*\|_n = O_p(r_n), \quad (8)$$

$$\|\hat{g}_n - f^*\|_P = O_p(r_n), \quad (9)$$

and, for every multi-index α with $0 \leq |\alpha| \leq m-1$,

$$\|D^\alpha(\hat{g}_n - f^*)\|_{L_2(P_X)} = O_p\left(n^{-(m-|\alpha|)/(2m+d)}\right). \quad (10)$$

The same conclusion holds if (7) is weakened to $J(\hat{g}_n) = O_p(1)$.

Proof. Let $\Delta_n = \hat{g}_n - f^*$. On the event (7), $\Delta_n \in \mathcal{H}_M$ after increasing M by a fixed amount depending on $J(f^*)$. Put

$$x_n = \|\Delta_n\|_n, \quad p = 1 - \frac{d}{2m} \in (0, 1).$$

By (2) and Lemma 2.2,

$$x_n^2 \leq 2\mathbb{P}_n \varepsilon \Delta_n \leq 2|\mathbb{P}_n \varepsilon \Delta_n| \leq 2Z_n \left(n^{-1/2} x_n + n^{-1/2} (x_n + r_n)^p \right)$$

with probability tending to one. Since $r_n^{2-p} = n^{-1/2}$, the elementary inequality

$$x^2 \leq A n^{-1/2} x + A n^{-1/2} (x + r_n)^p \implies x \leq C_A r_n$$

holds for every fixed $A < \infty$. Because $Z_n = O_p(1)$, this gives (8). Then (9) follows from (6).

Finally, the usual Sobolev interpolation inequality gives, for $k = |\alpha| \leq m-1$,

$$\|D^\alpha \Delta_n\|_{L_2(\Omega)} \leq C \|\Delta_n\|_{L_2(\Omega)}^{1-k/m} \|\Delta_n\|_{W^{m,2}(\Omega)}^{k/m}.$$

Because the density of X is bounded above and below, $\|\Delta_n\|_{L_2(\Omega)}$ and $\|\Delta_n\|_P$ are equivalent. Also

$$\|\Delta_n\|_{W^{m,2}(\Omega)} \leq C \{J(\hat{g}_n)^{1/2} + J(f^*)^{1/2} + \|\Delta_n\|_{L_2(\Omega)}\} = O_p(1).$$

Substituting (9) proves (10). $\square$

4 The exact oracle tolerance

Theorem 4.1 (Exact oracle tolerance). *Assume Assumption 2.1. Let*

$$S_n = V_n = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2.$$

Then f^ is feasible for Problem (1),*

$$J(\hat{g}_n) \leq J(f^*)$$

with probability tending to one, Lim's inequality (2) holds, and the rates (8)–(10) hold.

Proof. Because $E_n(f^*) = V_n = S_n$, the function f^* is feasible. By roughness minimality,

$$J(\hat{g}_n) \leq J(f^*).$$

The feasibility of $\hat{g}_n$ gives

$$\mathbb{P}_n(\varepsilon_i - \Delta_n(X_i))^2 \leq S_n = V_n,$$

where $\Delta_n = \hat{g}_n - f^*$. Expanding the square,

$$V_n - 2\mathbb{P}_n \varepsilon \Delta_n + \|\Delta_n\|_n^2 \leq V_n,$$

and hence (2) follows. Proposition 3.1 now gives the rates. $\square$

Remark 4.2. *Theorem 4.1 is the cleanest corrected statement. It is mathematically decisive but oracle: V_n uses $\varepsilon_i = Y_i - f^*(X_i)$ and therefore is not observable without extra information about f^* or the noise.*

5 A one-sided minimax band

The exact equality $S_n = V_n$ is not the only oracle condition. A small undershoot still forces (2). The nontrivial point is to show that the estimator does not become too rough while reducing the residual below V_n . The next assumption is one convenient way to make that correction argument completely explicit.

Assumption 5.1 (Projection setting for the undershoot lemma). *In addition to Assumption 2.1, assume that $\Omega = [0, 1]^d$, that X_i are uniform on Ω , and that the errors are independent $N(0, \sigma^2)$ variables. Let $\{\phi_j : j \geq 1\}$ be the tensor-product cosine basis of $L_2([0, 1]^d)$, ordered by increasing Laplacian eigenvalue. There is a bounded Sobolev extension operator from $W^{m,2}(\Omega)$ to $W^{m,2}(\mathbb{R}^d)$, so each ϕ_j may be regarded as the restriction of an element of $\mathcal{F}_m$ satisfying*

$$J(\phi_j) \leq Cj^{2m/d}, \quad \|\phi_j\|_\infty \leq C. \quad (11)$$

Lemma 5.2 (Feasible correction for a small undershoot). *Assume Assumption 5.1. If*

$$0 \leq \rho_n := V_n - S_n = O_p(r_n^2), \quad (12)$$

then

$$J(\hat{g}_n) = O_p(1). \quad (13)$$

Proof. It suffices to prove tightness. Fix $A < \infty$ and work on the event $\rho_n \leq Ar_n^2$. We shall construct a feasible competitor with roughness bounded by a constant depending on A only. At the end A is allowed to increase, which proves (13) because $\rho_n/r_n^2 = O_p(1)$.

Choose

$$N = \lceil KAnr_n^2 \rceil = \lceil KAn^{d/(2m+d)} \rceil,$$

where K is a large constant fixed below. Let Φ be the $n \times N$ matrix with entries $\Phi_{ij} = \phi_j(X_i)$ and let

$$G_n = \frac{1}{n} \Phi^T \Phi.$$

Because the basis functions are bounded and orthonormal in $L_2(P_X)$, matrix Bernstein's inequality gives

$$\mathbb{P}\{\lambda_{\min}(G_n) \geq 1/2, \lambda_{\max}(G_n) \leq 3/2\} \rightarrow 1, \quad (14)$$

using $N \log N/n \rightarrow 0$.

On the event (14), let q_n be the empirical least-squares projection of the noise vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T$ onto $\text{span}\{\phi_1, \dots, \phi_N\}$:

$$q_n(x) = \sum_{j=1}^N \hat{\beta}_j \phi_j(x), \quad \hat{\beta} = G_n^{-1} \frac{1}{n} \Phi^T \varepsilon.$$

Then

$$B_n := \mathbb{P}_n q_n^2 = \mathbb{P}_n \varepsilon q_n = \frac{1}{n} \varepsilon^T P_\Phi \varepsilon,$$

where P_Φ is the Euclidean orthogonal projector onto the column space of Φ . Conditional on the design and on (14), P_Φ has rank N , and therefore

$$\frac{nB_n}{\sigma^2} \sim \chi_N^2.$$

Thus, with probability tending to one,

$$B_n \geq \frac{\sigma^2}{4} \frac{N}{n} \geq \frac{\sigma^2 K}{8} Ar_n^2. \quad (15)$$

Choose K so large that the right side in (15) is at least $2Ar_n^2$. Then $B_n \geq 2\rho_n$ on the event under consideration.

It remains to bound the roughness of q_n . Let R_N be the $N \times N$ matrix with entries

$$(R_N)_{jk} = \sum_{|\alpha|=m} \int_{\mathbb{R}^d} D^\alpha \phi_j(x) D^\alpha \phi_k(x) dx.$$

Then $J(q_n) = \widehat{\beta}^T R_N \widehat{\beta}$. Conditional on the design and on (14),

$$\mathbb{E}\{J(q_n) \mid X_1, \dots, X_n\} = \frac{\sigma^2}{n} \text{tr}(R_N G_n^{-1}) \leq \frac{2\sigma^2}{n} \text{tr}(R_N) = \frac{2\sigma^2}{n} \sum_{j=1}^N J(\phi_j).$$

By (11),

$$\frac{1}{n} \sum_{j=1}^N J(\phi_j) \leq \frac{C}{n} N^{1+2m/d} \leq C_A,$$

because $N \asymp_A n^{d/(2m+d)}$. Markov's inequality gives $J(q_n) = O_p(1)$ on the event $\rho_n \leq Ar_n^2$.

Since $B_n \geq \rho_n$, choose $t_n \in [0, 1]$ satisfying

$$(2t_n - t_n^2)B_n = \rho_n.$$

The function $\widetilde{f}_n = f^* + t_n q_n$ is feasible, because

$$E_n(\widetilde{f}_n) = \mathbb{P}_n(\varepsilon - t_n q_n)^2 = V_n - (2t_n - t_n^2)B_n = S_n.$$

Consequently,

$$J(\widehat{g}_n) \leq J(\widetilde{f}_n) \leq 2J(f^*) + 2J(q_n) = O_p(1)$$

on the event $\rho_n \leq Ar_n^2$. Letting A increase proves the lemma. $\square$

Theorem 5.3 (One-sided oracle band). *Assume Assumption 5.1. If*

$$0 \leq V_n - S_n = O_p(r_n^2),$$

then Lim's inequality (2) holds with probability tending to one, and the rates (8)–(10) hold.

Proof. The upper inequality $S_n \leq V_n$ gives (2) by the same expansion used in the proof of Theorem 4.1. Lemma 5.2 gives $J(\widehat{g}_n) = O_p(1)$. Proposition 3.1 then gives the rates. $\square$

6 Why deterministic variance-level tolerances cannot do this

The preceding results are oracle results because they compare S_n with the realized noise energy V_n , not merely with its mean σ^2 . This distinction is unavoidable at the squared minimax scale.

Proposition 6.1 (A deterministic obstruction). *Assume that $\varepsilon_1, \varepsilon_2, \dots$ are iid, that $\mathbb{E}\varepsilon_1^2 = \sigma^2$, that $0 < \text{Var}(\varepsilon_1^2) < \infty$, and that $2m > d$. Let s_n be deterministic and satisfy*

$$s_n = \sigma^2 + O(r_n^2), \quad r_n = n^{-m/(2m+d)}.$$

Then

$$\mathbb{P}\{s_n \leq V_n\} \longrightarrow \frac{1}{2}$$

whenever $\sqrt{n}(s_n - \sigma^2) \rightarrow 0$; in particular no such deterministic tolerance can imply $s_n \leq V_n$ with probability tending to one.

Proof. Since $2m > d$,

$$r_n^2 = n^{-2m/(2m+d)} = o(n^{-1/2}).$$

Thus $\sqrt{n}(s_n - \sigma^2) \rightarrow 0$. The central limit theorem applied to $\varepsilon_i^2 - \sigma^2$ gives

$$\sqrt{n}(V_n - \sigma^2) \Rightarrow N(0, \text{Var}(\varepsilon_1^2)).$$

Therefore

$$\mathbb{P}\{s_n \leq V_n\} = \mathbb{P}\{\sqrt{n}(V_n - \sigma^2) \geq \sqrt{n}(s_n - \sigma^2)\} \longrightarrow \mathbb{P}\{Z \geq 0\} = \frac{1}{2}.$$

□

Remark 6.2. *A stronger almost-sure version follows from the law of the iterated logarithm: a deterministic tolerance that is below σ^2 by only $o(\sqrt{\log \log n/n})$ will still exceed V_n infinitely often almost surely. Since $r_n^2 = o(n^{-1/2})$, the squared minimax scale is much smaller than the intrinsic fluctuations of V_n .*

7 Conclusion

The repaired result is intentionally narrower than the originally proposed statement. The true rate-enabling sign condition is one-sided:

$$S_n \leq V_n.$$

The exact oracle choice $S_n = V_n$ gives a complete and rigorous solution to Lim's displayed condition (13), and hence to the empirical rate question for Problem (B). The slightly more general oracle band $0 \leq V_n - S_n = O_p(r_n^2)$ also gives the same rates under the projection assumptions of Section 5.

What remains open is the fully data-driven version in the ordinary model with one noisy observation per design point: construct an observable S_n that is both one-sided relative to V_n and accurate at the squared minimax scale. Proposition 6.1 shows that deterministic variance-level choices cannot do this. Additional information, such as replicated observations or an independent high-accuracy estimate of the realized noise energy, would be needed to turn the oracle condition into a practical tuning rule.

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